GEMM Analytical Model - Equations

Auto-generated from PYTHON_FORMULAS and LATEX_SYMBOL_OVERRIDES in gemmanalytics.py.

Output order: execution

row 22 - INPUT A BYTES PER ELEMENT

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matA})}]$$

row 23 - INPUT B BYTES PER ELEMENT

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matB})}]$$

row 25 - OUTPUT BYTES PER ELEMENT AFTER DOWN CONVERSION

$$\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{D}\downarrow)} = \mathrm{DataFormatToBytes}[\mathrm{Fmt}^{(\mathrm{matD}\downarrow)}]$$

row 33 - EU COUNT

$$|\mathrm{EU}| = |\mathrm{XeCore}|_{\mathrm{pXeCU}} \times |\mathrm{EU}|_{\mathrm{pXeCore}} \times |\mathrm{XeCU}|$$

row 102 - TOTAL L2 SIZE B FOR A SINGLE INSTANCE

$$|\mathrm{L2}|_{\mathrm{B}} = \mathrm{L2BankSize}_{\mathrm{MB}} \times |\mathrm{L2Banks}|_{\mathrm{pXeCU}} \times 1024 \times 1024$$

row 125 - MAX POSSIBLE HBM BW FREQ B CLK

$$\mathrm{MemBW}_{\mathrm{BpClk}}^{\mathrm{max}} = \frac{\mathrm{MemBW}_{\mathrm{GBps}}^{\mathrm{max}}}{\mathrm{f}_{\mathrm{GHz}}^{(\mathrm{GT})}}$$

row 36 - MMA MAC THROUGHPUT PER XECORE

$$\tau_{\mathrm{mMACpClk \cdot XeCore}}^{(\mathrm{peak})} = \min\left(\frac{4}{\frac{\mathrm{FLOOR}(\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})} \times 2, 1)}{2}}, \frac{4}{\frac{\mathrm{FLOOR}(\mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})} \times 2, 1)}{2}}\right) \times \mathrm{D}_{\mathrm{DPAS}} \times 16 \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 42 - CLKS PER DPAS

$$\mathrm{CLKS}_{\mathrm{DPAS}} = \frac{\mathrm{M}_{\mathrm{pThread}} \times \mathrm{K}_{\mathrm{pThread}} \times \mathrm{N}_{\mathrm{pThread}}}{\frac{\tau_{\mathrm{mMACpClk \cdot XeCore}}^{(\mathrm{peak})}}{|\mathrm{EU}|_{\mathrm{pXeCore}}}}$$

row 13 - WAVES

$$\mathrm{N}_{\mathrm{waves}} = \mathrm{CEILING}(|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})}, 1) \times \mathrm{CEILING}(|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})}, 1)$$

row 14 - TG TILES IN N

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{TG})} = \frac{\mathrm{N}_{\mathrm{dim}}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}}$$

row 15 - TG TILES IN M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{TG})} = \frac{\mathrm{M_{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}}$$

row 16 - TG CLUSTER TILES IN N

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{TG\ Cluster})} = \frac{\mathrm{N_{dim}}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCoreCluster})}}$$

row 17 - TG CLUSTER TILES IN M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{TG\ Cluster})} = \frac{\mathrm{M_{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCoreCluster})}}$$

row 18 - XECU TILES IN N

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{XeCU})} = \frac{\mathrm{N_{dim}}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}$$

row 19 - XECU TILES IN M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{XeCU})} = \frac{\mathrm{M_{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}$$

row 20 - GPU TILES IN N

$$|\mathrm{Tiles}|_{\mathrm{N}}^{(\mathrm{GPU})} = \frac{\frac{\mathrm{N_{dim}}}{\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}}{\mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 21 - GPU TILES IN M

$$|\mathrm{Tiles}|_{\mathrm{M}}^{(\mathrm{GPU})} = \frac{\frac{\mathrm{M_{dim}}}{\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}}}{\mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 43 - THREAD WIDTH IN UNITS OF ELEMENTS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})} = \mathrm{N_{pThread}}$$

row 44 - THREAD HEIGHT IN UNITS OF ELEMENTS

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})} = \mathrm{M_{pThread}}$$

row 47 - TG WIDTH IN UNITS OF ELEMENT REALIZED BY MULTIPLE MMA ITERATIONS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})} = \mathrm{W}_{\mathrm{thread}}^{(\mathrm{ThreadGroup})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{Thread})}$$

row 48 - TG HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})} = \mathrm{H}_{\mathrm{thread}}^{(\mathrm{ThreadGroup})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{Thread})}$$

row 52 - XECORE CLUSTER WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCoreCluster})} = \mathrm{W}_{\mathrm{TG,keep\ cluster\ size\ as\ 4}}^{(\mathrm{XeCoreCluster})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}$$

row 53 - XECORE CLUSTER HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCoreCluster})} = \mathrm{H}_{\mathrm{TG,keep\ cluster\ size\ as\ 4}}^{(\mathrm{XeCoreCluster})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}$$

row 56 - XECU TILE WIDTH IN UNITS OF ELEMENT

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})} = \mathrm{W}_{\mathrm{TG}}^{(\mathrm{XECUTile})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}$$

row 57 - XECU TILE HEIGHT IN UNITS OF ELEMENT

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})} = \mathrm{H}_{\mathrm{TG}}^{(\mathrm{XECUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{ThreadGroup})}$$

row 59 - GPU TILE HEIGHT IN XECU UINT

$$\mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} = \frac{|\mathrm{XeCU}|}{\mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})}}$$

row 60 - GPU TILE WIDTH IN UNITS OF ELEMETNS

$$\mathrm{W}_{\mathrm{element}}^{(\mathrm{GPUTile})} = \mathrm{W}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} \times \mathrm{W}_{\mathrm{element}}^{(\mathrm{XeCUTile})}$$

row 61 - GPU TILE HEIGHT IN UNITS OF ELEMETNS

$$\mathrm{H}_{\mathrm{element}}^{(\mathrm{GPUTile})} = \mathrm{H}_{\mathrm{XeCU}}^{(\mathrm{GPUTile})} \times \mathrm{H}_{\mathrm{element}}^{(\mathrm{XeCUTile})}$$

row 63 - MAT A INPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matA})} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{K}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matA})}$$

row 64 - MAT B INPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matB})} = \mathrm{K}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{matB})}$$

row 65 - MAT C INPUT D OUTPUT SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matC,matD})} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Bytes}_{\mathrm{pElement}}^{(\mathrm{D}\downarrow)}$$

row 66 - MAT D INTERMEDIATE SIZE B

$$\mathrm{Size}_{\mathrm{B}}^{(\mathrm{matD}\downarrow)} = \mathrm{M}_{\mathrm{dim}} \times \mathrm{N}_{\mathrm{dim}} \times \mathrm{Byte}_{\mathrm{pElement}}^{(\mathrm{matD})}$$

row 103 - WORKING DATA SET SIZE OF K IN L2 CORRESP 20K CLOCKS OF THREAD DIVERGENCE

$$|\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}} = \min(20000 \times \frac{\mathrm{K_{pThread}}}{\mathrm{CLKS_{DPAS}}}, \mathrm{K_{dim}})$$

row 104 - TOTAL REQUIRED L2 SIZE FOR IDEAL HIT RATE B FOR A SINGLE INSTANCE AND SINGLE WAVE

$$\begin{aligned}\mathrm{TotalRequiredL2Size}^{\mathrm{1\ instance,\ 1\ wave}}_{\mathrm{matB\ alwayshit}} &= \mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}} \times \mathrm{Bytes}^{(\mathrm{matA})}_{\mathrm{pElement}} \times |\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}} \\ &+ \mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}} \times \mathrm{Bytes}^{(\mathrm{matB})}_{\mathrm{pElement}} \times |\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}} \\ &+ \mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}} \times \mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}} \times \mathrm{Bytes}^{(\mathrm{D}\downarrow)}_{\mathrm{pElement}}\end{aligned}$$

row 37 - CLK SPECIFIED EFFICIENCY

$$\mathrm{T}^{(\mathrm{total})}_{\mathrm{clk}} = \frac{\left(\mathrm{T}^{(\mathrm{total})}_{\mathrm{clk}}\right)_{\mathrm{num}}}{\eta_{\mathrm{systolic}}}$$

$$\left(\mathrm{T}^{(\mathrm{total})}_{\mathrm{clk}}\right)_{\mathrm{num}} = \frac{\left(\left(\mathrm{T}^{(\mathrm{total})}_{\mathrm{clk}}\right)_{\mathrm{num}}\right)_{\mathrm{num}}}{\tau^{(\mathrm{peak})}_{\mathrm{mMACpClk\cdot XeCore}}}$$

$$\left(\left(\mathrm{T}^{(\mathrm{total})}_{\mathrm{clk}}\right)_{\mathrm{num}}\right)_{\mathrm{num}} = \frac{\frac{\mathrm{M_{dim}} \times \mathrm{K_{dim}} \times \mathrm{N_{dim}}}{|\mathrm{XeCore}|_{\mathrm{pXeCU}}}}{|\mathrm{XeCU}|}$$

row 69 - TOTAL L2 READ B

$$\begin{aligned}\mathrm{L2Rd}^{(\mathrm{total})}_{\mathrm{B}} &= \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \mathrm{CEILING}(\frac{\mathrm{N_{dim}}}{\mathrm{W}^{(\mathrm{ThreadGroup,realized\ by\ multiple\ MMA\ iterations})}_{\mathrm{element}}}, 1) \\ &+ \mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \mathrm{CEILING}(\frac{\mathrm{M_{dim}}}{\mathrm{H}^{(\mathrm{ThreadGroup})}_{\mathrm{element}}}, 1)\end{aligned}$$

row 70 - TOTAL L2 WRITE B

$$\mathrm{L2Wr}^{(\mathrm{total})}_{\mathrm{B}} = \mathrm{Size}^{(\mathrm{matC,matD})}_{\mathrm{B}}$$

row 71 - TOTAL L1 READ B

$$\begin{aligned}\mathrm{L1Rd}^{(\mathrm{total})}_{\mathrm{B}} &= \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \mathrm{CEILING}(\frac{\mathrm{N_{dim}}}{\mathrm{W}^{(\mathrm{Thread})}_{\mathrm{element}}}, 1) \\ &+ \mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \mathrm{CEILING}(\frac{\mathrm{M_{dim}}}{\mathrm{H}^{(\mathrm{Thread})}_{\mathrm{element}}}, 1)\end{aligned}$$

row 72 - TOTAL L1 WRITE B

$$\mathrm{L1Wr}_{\mathrm{B}}^{(\mathrm{total})} = \mathrm{Size}_{\mathrm{B}}^{(\mathrm{matC,matD})}$$

row 74 - L2 READ B XECORE CLK

$$\mathrm{L2RdB}_{\mathrm{p(Clk\cdot XeCore)}} = \frac{\frac{\mathrm{L2Rd}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{pXeCU}}}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 75 - L2 WRITE B XECORE CLK

$$\mathrm{L2WrB}_{\mathrm{p(Clk\cdot XeCore)}} = \frac{\frac{\mathrm{L2Wr}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{XeCU}| \times |\mathrm{XeCore}|_{\mathrm{pXeCU}}}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 76 - L2 READ WRITE B XECORE CLK

$$\begin{aligned}\mathrm{L2RdWrB}_{\mathrm{p(Clk\cdot XeCore)}} &= \mathrm{L2RdB}_{\mathrm{p(Clk\cdot XeCore)}} \\ &+ \mathrm{L2WrB}_{\mathrm{p(Clk\cdot XeCore)}}\end{aligned}$$

row 77 - L1 READ B XECORE CLK

$$\mathrm{L1RdB}_{\mathrm{p(Clk\cdot XeCore)}} = \mathrm{L1RdB}_{\mathrm{p(Clk\cdot EU)}} \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 78 - L1 WRITE B XECORECLK

$$\mathrm{L1WrB}_{\mathrm{p(Clk\cdot XeCore)}} = \mathrm{L1WrB}_{\mathrm{p(Clk\cdot EU)}} \times |\mathrm{EU}|_{\mathrm{pXeCore}}$$

row 79 - L1 READ B EU CLK

$$\mathrm{L1RdB}_{\mathrm{p(Clk\cdot EU)}} = \frac{\frac{\mathrm{L1Rd}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{EU}|}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 80 - L1 WRITE B EU CLK

$$\mathrm{L1WrB}_{\mathrm{p(Clk\cdot EU)}} = \frac{\frac{\mathrm{L1Wr}_{\mathrm{B}}^{(\mathrm{total})}}{|\mathrm{EU}|}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 83 - L2 READ MAX B XECORE CLK

$$\mathrm{L2RdB}_{\mathrm{p(Clk\cdot XeCore)}}^{(\mathrm{max})} = \frac{|\mathrm{XeCore}|_{\mathrm{pXeCU}} \times 64}{|\mathrm{XeCore}|_{\mathrm{pXeCU}}}$$

row 84 - L2 WRITE MAX B XECORE CLK

$$\mathrm{L2WrB}_{\mathrm{p(Clk\cdot XeCore)}}^{(\mathrm{max})} = \mathrm{L2RdB}_{\mathrm{p(Clk\cdot XeCore)}}^{(\mathrm{max})}$$

row 85 - L2 READ WRITE MAX B XECORE CLK

$$\mathrm{L2RdWrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}} = \mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}$$

row 90 - L2 READ B XECORE CLK PCT

$$\eta_{\mathrm{L2RdB/p(Clk\cdot XeCore)}} = \frac{\mathrm{L2RdB}_{\mathrm{p(Clk\cdot XeCore)}}}{\mathrm{L2RdB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}}$$

row 91 - L2 WRITE B XECORE CLK PCT

$$\eta_{\mathrm{L2WrB/p(Clk\cdot XeCore)}} = \frac{\mathrm{L2WrB}_{\mathrm{p(Clk\cdot XeCore)}}}{\mathrm{L2WrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}}$$

row 92 - L2 READ WRITE B XECORE CLK PCT

$$\eta_{\mathrm{L2RdWrB/p(Clk\cdot XeCore)}} = \frac{\mathrm{L2RdWrB}_{\mathrm{p(Clk\cdot XeCore)}}}{\mathrm{L2RdWrB}^{(\mathrm{max})}_{\mathrm{p(Clk\cdot XeCore)}}}$$

row 93 - L1 READ B EU CLK PCT

$$\eta_{\mathrm{L1RdB/p(Clk\cdot EU)}} = \frac{\mathrm{L1RdB}_{\mathrm{p(Clk\cdot EU)}}}{\mathrm{L1RdBW}^{\mathrm{max}}_{\mathrm{BpClk\cdot EU}}}$$

row 94 - L1 WRITE B EU CLK PCT

$$\eta_{\mathrm{L1WrB/p(Clk\cdot EU)}} = \frac{\mathrm{L1WrB}_{\mathrm{p(Clk\cdot EU)}}}{\mathrm{L1WrBW}^{\mathrm{max}}_{\mathrm{BpClk\cdot EU}}}$$

row 106 - L2 HIT RATE ASSUMED RANDOM ACCESS WITHIN THE WORKING DATA SET PCT

$$\mathrm{L2HitRate}_{\mathrm{random\ WS\ access}} = \min\left(\frac{|\mathrm{L2}|_{\mathrm{B}}}{\mathrm{TotalRequiredL2Size}^{\mathrm{1\ instance,\ 1\ wave}}_{\mathrm{matB\ alwayshit}}}, 1\right)$$

row 107 - L2 MISS RATE PCT

$$\mathrm{L2MissRate} = 1 - \mathrm{L2HitRate}_{\mathrm{random\ WS\ access}}$$

row 108 - TOTAL L2 READ TRAFFIC B

$$\mathrm{L2RdB}_{\mathrm{total}} = \mathrm{L2Rd}^{(\mathrm{total})}_{\mathrm{B}}$$

row 110 - PROBABILITY OF MATA HIT IN L2 DURING A NON FIRST WAVE

PCT

$$
\begin{aligned}
\mathrm{P}^{(\mathrm{matA})}_{\mathrm{L2Hit,\ after\ 1st\ wave}} = {} & \mathrm{IF}(\mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \\
& +\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \\
& +\mathrm{Size}^{(\mathrm{matC,matD})}_{\mathrm{B}} \leq |\mathrm{L2}|_{\mathrm{B}}, 1.0, \mathrm{IF}(\mathrm{K}_{\mathrm{dim}} > 2 \times |\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}}, 0.0, 1 \\
& -\frac{\mathrm{K}_{\mathrm{dim}} - |\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}}}{|\mathrm{WorkingSet}|^{(\mathrm{K\ in\ L2})}_{\mathrm{20K\ clks\ of\ thread\ divergence}}}))
\end{aligned}
$$

row 111 - PROBABILITY OF MATB HIT IN L2 DURING A NON FIRST WAVE

PCT

$$
\begin{aligned}
\mathrm{P}^{(\mathrm{matB})}_{\mathrm{L2Hit,\ after\ 1st\ wave}} = {} & \mathrm{IF}(\mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \\
& +\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \\
& +\mathrm{Size}^{(\mathrm{matC,matD})}_{\mathrm{B}} \leq |\mathrm{L2}|_{\mathrm{B}}, 1.0, 0.0)
\end{aligned}
$$

row 112 - PROBABILITY OF MATA MISS IN L2 DURING A NON FIRST WAVE

PCT

$$
\begin{aligned}
\mathrm{P}^{(\mathrm{matA})}_{\mathrm{L2Miss,\ after\ 1st\ wave}} = {} & 1 \\
& -\mathrm{P}^{(\mathrm{matA})}_{\mathrm{L2Hit,\ after\ 1st\ wave}}
\end{aligned}
$$

row 113 - PROBABILITY OF MATB MISS IN L2 DURING A NON FIRST WAVE

PCT

$$
\begin{aligned}
\mathrm{P}^{(\mathrm{matB})}_{\mathrm{L2Miss,\ after\ 1st\ wave}} = {} & 1 \\
& -\mathrm{P}^{(\mathrm{matB})}_{\mathrm{L2Hit,\ after\ 1st\ wave}}
\end{aligned}
$$

row 116 - TOTAL HBM READ B AFTER A COMPLETION OF A WAVE CONSIDER

COLD CACHE

$$
\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}} = \frac{\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}\right)_{\mathrm{num}}}{2}
$$

$$\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}\right)_{\mathrm{num}} = \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{N_{dim}}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matA})}_{\mathrm{L2Miss,\ after\ 1st\ wave}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{M_{dim}}}{\mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matB})}_{\mathrm{L2Miss,\ after\ 1st\ wave}}$$
$$+\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}} \times \mathrm{L2MissRate}$$

$$\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}} = \mathrm{L2RdB_{total}}$$
$$-\left(\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}}\right)_{\mathrm{aux1}}$$

$$\left(\left(\left(\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}\right)_{\mathrm{num}}\right)_{\mathrm{aux4}}\right)_{\mathrm{aux1}} = \mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matA})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{N_{dim}}}{\mathrm{W}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matA})}_{\mathrm{L2Miss,\ after\ 1st\ wave}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}}$$
$$+\mathrm{Size}^{(\mathrm{matB})}_{\mathrm{B}} \times \left(\mathrm{CEILING}(\frac{\mathrm{M_{dim}}}{\mathrm{H}^{(\mathrm{XeCUTile})}_{\mathrm{element}}}, 1) - 1\right) \times \mathrm{P}^{(\mathrm{matB})}_{\mathrm{L2Miss,\ after\ 1st\ wave}}$$

row 117 - TOTAL HBM WRITE B

$$\mathrm{MemWrB}^{(\mathrm{total})} = \mathrm{Size}^{(\mathrm{matC,matD})}_{\mathrm{B}}$$

row 118 - TOTAL HBM B

$$\mathrm{MemRdWrB}^{(\mathrm{total})} = \mathrm{MemWrB}^{(\mathrm{total})} + \mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}$$

row 121 - HBM READ B CLK

$$\mathrm{MemRdB_{pClk}} = \frac{\mathrm{MemRdB}^{(\mathrm{wave})}_{\mathrm{cold\ start}}}{\mathrm{T}^{(\mathrm{total})}_{\mathrm{clk}}}$$

row 122 - HBM WRITE B CLK

$$\mathrm{MemWrB}_{\mathrm{pClk}} = \frac{\mathrm{MemWrB}^{(\mathrm{total})}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 123 - HBM TOTAL B CLK

$$\mathrm{MemRdWrB}_{\mathrm{pClk}} = \frac{\mathrm{MemRdWrB}^{(\mathrm{total})}}{\mathrm{T}_{\mathrm{clk}}^{(\mathrm{total})}}$$

row 126 - HBM BW PCT

$$\eta_{\mathrm{MemBW}} = \frac{\mathrm{MemRdWrB}_{\mathrm{pClk}}}{\mathrm{MemBW}_{\mathrm{BpClk}}^{\mathrm{max}}}$$

row 127 - GTI READ BW PCT

$$\eta_{\mathrm{GTIRdBW}} = \frac{\mathrm{MemRdB}_{\mathrm{pClk}}}{\mathrm{GtiRdBW}_{\mathrm{BpClk}}^{\mathrm{max}}}$$

row 128 - GTI WRITE BW PCT

$$\eta_{\mathrm{GTIWrBW}} = \frac{\mathrm{MemWrB}_{\mathrm{pClk}}}{\mathrm{GtiWrBW}_{\mathrm{BpClk}}^{\mathrm{max}}}$$